Functions, Modulus and Quadratics

Section 1: Function Mapping, Domain/Range and Inverses

1 The functions f and g are defined for $x \in \mathbf{R}$ by $f(x) = 2x - 5$ and $g(x) = x^2 + 3$. Find the composite value $(g \circ f)(4)$.

a. 12

b. 21

c. 14

d. 9

2 A function is defined by $f(x) = \frac{3x+1}{x-2}$ for $x \neq 2$. Find an expression for the inverse function $f^{-1}(x)$ and state its domain.

a. $f^{-1}(x) = \frac{2x+1}{x-3}, \quad x \neq 3$

b. $f^{-1}(x) = \frac{x-2}{3x+1}, \quad x \neq -\frac{1}{3}$

c. $f^{-1}(x) = \frac{2x-1}{x+3}, \quad x \neq -3$

d. $f^{-1}(x) = \frac{3x-2}{x-1}, \quad x \neq 1$

3 The square root function $h(x) = 4 - \sqrt{2x - 6}$ is defined on its maximal real domain. Determine the domain and range of h.

a. Domain: $x \geq 3$, Range: $h(x) \leq 4$

b. Domain: $x \geq 3$, Range: $h(x) \geq 4$

c. Domain: $x > 3$, Range: $h(x) \in \mathbf{R}$

d. Domain: $x \leq 3$, Range: $h(x) \leq 0$

Section 2: Modulus Equations and Inequalities

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4 Solve the linear modulus equation $|3x - 4| = 2x + 1$ for all valid real solutions.

a. $x = 5$ and $x = \frac{3}{5}$

b. $x = 5$ only

c. $x = -1$ and $x = 3$

d. No real solutions

5 Find the complete solution set for the absolute value inequality $|2x - 7| < 5$.

a. $1 < x < 6$

b. $x < 1$ or $x > 6$

c. $-6 < x < 1$

d. $2 < x < 5$

6 Solve the two-modulus equation $|x + 2| = |2x - 5|$.

a. $x = 7$ and $x = 1$

b. $x = 7$ and $x = -1$

c. $x = 3$ and $x = -3$

d. $x = \frac{7}{3}$ only

Section 3: Quadratic Discriminant and Line-Parabola Intersections

7 Find the set of values of k for which the quadratic equation $x^2 + (k + 2)x + (2k + 1) = 0$ has two distinct real roots.

a. $k < 0$ or $k > 4$

b. $0 < k < 4$

c. $k \leq 0$ or $k \geq 4$

d. $-4 < k < 0$

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8 The line $y = 2x + c$ is tangent to the parabola $y = x^2 - 4x + 7$. Determine the exact value of the constant c.

a. $c = -2$

b. $c = 2$

c. $c = -7$

d. $c = 3$

9 Find the range of values of m for which the line $y = mx - 1$ does not intersect the curve $y = 2x^2 + 3x + 1$.

a. $-1 < m < 7$

b. $m < -1$ or $m > 7$

c. $-7 < m < 1$

d. $m < 3$ or $m > 5$

Section 4: Completing the Square and Applied Optimization

10 By completing the square, express $f(x) = 2x^2 - 12x + 23$ in the form $a(x - h)^2 + k$, and state the minimum value of $f(x)$.

a. $2(x - 3)^2 + 5$, minimum value 5

b. $2(x - 3)^2 + 23$, minimum value 23

c. $2(x - 6)^2 - 13$, minimum value -13

d. $(2x - 6)^2 + 5$, minimum value 5

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11 A rectangular field has three sides fenced with 80 m of fencing against a river. Find the maximum possible area.

a. $800\ \text{m}^2$

b. $400\ \text{m}^2$

c. $1600\ \text{m}^2$

d. $600\ \text{m}^2$

12 Find all real solutions to the biquadratic equation $x^4 - 10x^2 + 9 = 0$.

a. $x = \pm 1$ and $x = \pm 3$

b. $x = 1$ and $x = 9$

c. $x = \pm 1$ only

d. $x = \pm\sqrt{10}$

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